SOHOM CHATTERJEE

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CS undergraduate building Gen AI systems in production, not just coursework. At Lumeo I'm building an AI-native product from scratch with LangChain and LangGraph agentic pipelines. At HackQuest I built an ML pipeline that scores user progress. On campus I teach ML to 15+ teams and run AI workshops. Same throughline everywhere: build the model, ship it, then teach someone else how it works. Also have two open-source fixes merged or filed in developer-tooling and AI-eval codebases.

OPEN SOURCE CONTRIBUTIONS

- **Graphify (Graphify-Labs/graphify)**: Found and fixed a performance bug in the TypeScript import-type normalizer. It was scanning every type-argument range on each match, which pinned CPU at 100% on large files. Rewrote it as a sorted-index lookup with the same output. Merged in v0.9.56 (issue #3359, commit 67f99bd).
- **UKGovernmentBEIS/inspect_ai**: Found three more places where Inspect AI's grading and multiple-choice templates used raw str.format() and broke on unescaped JSON in examples. Filed the issue and wrote the fix, building on an earlier PR (issue #5276, stacked on PR #5256).

GEN AI, RAG & AGENTIC AI PROJECTS

- **Archi.dev** (TypeScript, Next.js, LangChain, Graph Reasoning): AI tool that plans microservice architecture by having an LLM reason over a graph of system components and their dependencies. GitHub · Live
- **IRIS Protocol** (Python, LangChain, LangGraph, FAISS, Solana): agentic ML layer for an autonomous trading system. Four strategy agents (momentum, mean-reversion, breakout, adaptive research) run on a shared LangGraph state graph, use FAISS for retrieval, and log to a hash-committed ledger so performance is verifiable.
- **Reforge** (Python): ML system that retrains and restructures its own models instead of needing manual retraining. GitHub
- **Neural Network From Scratch** (Python, NumPy): feedforward network and backpropagation written from raw math, no ML framework, to understand what's actually happening under the hood. GitHub
- **Sage AI** (Python, Gemini 2.5 Flash, Mistral 7B, TTS): voice assistant that switches between Gemini and a Mistral 7B model I fine-tuned locally, depending on the task. GitHub
- **Shortlist AI** (Next.js, NLP, AWS S3, DynamoDB): serverless tool that scores resumes against job descriptions with an NLP pipeline. GitHub
- **XAI for Plant Disease Detection** (CNN, SHAP, LIME, Grad-CAM++, TensorFlow): co-authoring a paper on explainable CNNs for plant disease detection, with a 20-paper literature review and a visual explanation pipeline farmers can actually read.
- **Brier** (Solidity, zkML): on-chain insurance that slashes AI model operators for being confidently wrong, not just wrong, using zkML to prove the model's stated confidence. GitHub
- **DACAP** (Solidity, Rust, AI Agents): AI auditor that checks, optimizes, and upgrades smart contracts for autonomous finance systems. GitHub

EXPERIENCE

Lumeo, CEO & Co-founder | Kolkata **Nov. 2025 - Present**

- Leading a 4-person team building an AI-native financial OS: NestJS/Prisma backend, Next.js 14 frontend, a Python ML server, and Stellar/Soroban smart contracts. 40,000+ people on the waitlist without any ad spend.
- Own the product architecture end-to-end, including the AI-driven compliance and reconciliation layer that's the core of the product.

HackQuest India, Developer Advocate (Internship) | Remote **Nov. 2025 - Present**

- Built an ML pipeline that scores user progress from in-platform activity, improving point-attribution accuracy by 50%. Mentored 25+ students on Web3 and Solidity development, and write technical tutorials for a global developer audience.

GeeksforGeeks SNU Chapter, Technical Executive | Kolkata · Hybrid **Mar. 2025 - Nov. 2025**

- Mentored 15+ teams through full ML projects, from problem framing to deployment, using Python, TensorFlow, Flask, and Docker. Ran code reviews and hyperparameter-tuning sessions. Helped ship 10+ live demos.

Google, Gemini Student Campus Ambassador (Part-time) | India · Remote **Aug. 2025 - Jun. 2026**

- Lead AI-driven initiatives on campus: organize hands-on workshops and tech events, and build a community of student developers around emerging AI tools.

LNC Community, Dev Team Member | Remote **Oct. 2025 - Present**

- Contribute to and maintain open-source projects in the LNC Network. Work with designers and other developers to ship features and solve real development problems.

EDUCATION

Sister Nivedita University (MAKAUT): B.Tech, Computer Science & Engineering (Data Science) **Aug. 2024 - 2028**

- Coursework: Data Structures & Algorithms, Machine Learning, DBMS, Operating Systems, OOP. Strong focus on algorithmic problem-solving.

TECHNICAL SKILLS

- **Gen AI / ML:** LangChain, LangGraph, RAG pipelines, agentic AI workflows, LLMs (Gemini, Mistral 7B), HuggingFace, TensorFlow, PyTorch, scikit-learn, model fine-tuning & inference, SHAP/LIME/Grad-CAM++, NLP, MLOps
- **Core CS:** Data Structures & Algorithms, System Design, Python, TypeScript, JavaScript (Node.js), SQL
- **Backend & Cloud:** NestJS, FastAPI, Flask, PostgreSQL, Prisma, Redis, Docker, REST API integration, AWS, GCP, Vercel

CERTIFICATIONS & ACHIEVEMENTS

- SIH'24 Finalist (Smart India Hackathon, national) · Google Solution Challenge, Certificate of Achievement (2025)
- MLOps for Generative AI (Google Cloud Skills Boost) · Transformer Models & BERT (Google Cloud Skills Boost) · Full Stack & AI/ML Developer (Sheryians Coding School)